

KANG-TZU LIU

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Experience

Foxconn Industrial Internet

March 2024 – Present

Software Engineer

Houston, Texas

- Built and automated CI/CD pipelines in **Azure DevOps** to deploy a Next.js application to Windows servers using IIS for hosting and NSSM for service management.
- Led the migration of a legacy manufacturing control system from ASP.NET Web Forms to a modern web architecture using **NextJS** and **C#**, ensuring minimal downtime and seamless adoption by factory users.
- Designed and developed **RESTful APIs** using **C#** for the factory's manufacturing system, including endpoints for real-time production data tracking and inventory management, handling over **5,000 requests** per day.

Cocotrees

Dec 2023 – March 2024

Software Engineer Intern

Remote

- Designed a geographic information system(GIS) that integrated **Google Map Platform API** to visualize spatial data, including slope distribution, vegetation coverage, and biodiversity. Customized clicking events and information windows for each object on the map, providing an interactive user experience.
- Developed full-stack in **NextJS** and **TypeScript**; Integrated **Prisma** and **MySQL** from AWS RDS for database migrations and optimization of data fetching processes.

Winbond Electronics

July 2023 – Aug 2023

Software Engineer Intern

Taichung, Taiwan

- Designed a manufacturing execution system using **Angular** and **JavaScript**, facilitating real-time monitoring of wafer status and the ability to manage wafer handling throughout the manufacturing process.
- Integrated **Oracle SQL** with **.NET** framework to retrieve and manipulate wafer information data.
- Implemented **UiPath Studio**, a Robotic Process Automation(RPA) application, to conduct automated stability and speed tests on various manufacturing system workflows, resulting in a **30% enhancement** in API efficiency.

Technical Skills

Language:	Python, JavaScript, TypeScript, C#, SQL, C++, C, , HTML/CSS
Technologies:	AWS, GCP, Heroku, Kubernetes, Docker, SQL Server, MySQL, PostgreSQL, MongoDB, Git, Jest
Tools:	React, Next.js, Node.js, Express.js, React Native, Prisma, Selenium, Tailwind CSS, UiPath Studio

Certifications

AWS Certified Cloud Practitioner

May 2025

Projects

ReliefAI: AI-Driven Room Styling (In Development - Co-Developer) | *React Native, Python* June 2025 - Present

- Building a mobile app that allows users to upload photos of rooms and receive **AI-generated** design styles along with furniture recommendations sourced from shopping platforms.
- Integrated **Stability API** for image generation, and **SerpAPI** for product search and recommendation automation.
- Developed frontend in **React Native** and backend in **Flask** to support AI-driven product recommendation flows.

Reliable File Transfer Protocol | *C*

March 2023

- Designed a file transfer protocol based on UDP that can transfer files in any size using only 1MB memory of buffer.
- Simulated **TCP** by implementing data transfer reliability under five severe Internet circumstances (e.g. packets out of order, duplicate, drop, delay, and data corruptions) by designing sliding window protocol, timeout re-transmission mechanism, and error detection with 1's complement checksum.

Ricebook | *JavaScript, React, Node.js, MongoDB, Heroku, AWS*

Dec 2022

- Implemented CI/CD pipelines with **Github Actions** to build **Docker images** then automatically deploy the backend to **AWS EC2 instances**.
- Designed and implemented a social networking platform that integrated the function of user registration, authentication, posting and commenting on articles, and uploading images.
- Developed full-stack in **JavaScript**, including **React** and **Redux**; Utilized **RESTful APIs** for client-server communication by Node.js, Express.js, MongoDB, and Mongoose

Education

Rice University

Aug 2022 – Dec 2023

Master of Computer Science

Houston, Texas

National Cheng Kung University

Sept 2017 – Jan 2022

B.S. in Computer Science and Information Engineering

Tainan, Taiwan